

A
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ON
“Design and Implementation of a Mobile-Based Emergency Alert System for Public
Safety”

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ABSTRACT

Personal safety and emergency response have become major concerns in today's society. Traditional emergency systems often suffer from delayed communication, lack of real-time location sharing, and limited disaster awareness capabilities. To address these challenges, the proposed system, CareWave: Design and Implementation of a Mobile-Based Emergency Alert System for Public Safety, provides an intelligent platform for emergency assistance, live location tracking, safe-zone monitoring, and disaster alert management.

The system enables users to instantly trigger SOS alerts, notify emergency contacts, share live location information, and receive real-time disaster notifications based on their geographical location. The application integrates GPS services, OpenStreetMap, Firebase Cloud Messaging (FCM), and secure backend services developed using Spring Boot. Additional features include geofencing, guardian tracking, emergency contact management, disaster monitoring, and notification history management. The backend utilizes Spring Security and JWT authentication to ensure secure access to application resources, while disaster monitoring services continuously analyze information from external APIs and generate alerts for users located within affected regions.

The geofencing module employs location-based monitoring techniques to detect entry and exit events from predefined safe zones. The proposed system improves emergency response efficiency, enhances user safety, and provides a reliable platform for personal protection and disaster awareness. The solution demonstrates the practical use of mobile computing, location-based services, cloud technologies, and real-time notification systems in public safety applications.

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LIST OF ABBREVIATIONS

GPS	Global Positioning System
JWT	JSON Web Token
FCM	Firebase Cloud Messaging
API	Application Programming Interface
REST	Representational State Transfer
SRS	Software Requirement Specification
UML	Unified Modeling Language
DBMS	Database Management System
SQL	Structured Query Language
UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 Background

Personal safety and emergency response have become major concerns in modern society due to increasing incidents of accidents, crimes, medical emergencies, and natural disasters. Traditional emergency response systems often rely on manual communication methods, causing delays in providing assistance during critical situations improving response times and enhancing public safety [1, 5].

The rapid growth of smartphone technology and location-based services has enabled the development of intelligent emergency management systems capable of improving response times and enhancing public safety.

CareWave is designed to address these challenges by providing an integrated emergency alert platform that combines real-time location tracking, emergency communication, geofencing, and disaster monitoring capabilities.

1.2 Problem Statement

Existing emergency response systems suffer from several limitations:

- Delayed emergency communication.
- Lack of real-time location sharing.
- Limited disaster awareness mechanisms.
- Dependence on manual intervention.
- Absence of intelligent safety monitoring.
- Inefficient emergency contact notification.

These issues often reduce the effectiveness of emergency response operations and increase risk for affected individuals.

1.3 Proposed Solution

CareWave is a mobile-based emergency alert and public safety platform that enables users to trigger SOS alerts, share live locations, monitor safe zones, receive disaster notifications, and communicate with emergency contacts.

The system utilizes GPS technology, OpenStreetMap services, Firebase Cloud Messaging, Spring Boot backend services, and PostgreSQL database management to provide a reliable emergency response infrastructure.

1.4 Objectives

1. Provide instant emergency alert generation.
2. Enable real-time location tracking.
3. Support emergency contact management.
4. Implement geofencing for safe-zone monitoring.
5. Provide disaster awareness notifications.
6. Maintain emergency event history.
7. Ensure secure user authentication.
8. Improve public safety and emergency response efficiency.

1.5 Scope of the Project

The system supports:

- User Registration and Authentication
- SOS Emergency Alerts
- Live Location Tracking
- Emergency Contact Management
- Guardian Tracking
- Geofencing and Safe-Zone Monitoring
- Disaster Monitoring
- Notification History Management
- Real-Time Push Notifications

1.6 System Features

Major features of CareWave include:

- User Authentication and Authorization
- Emergency SOS Alert System
- Live Location Tracking
- Emergency Contact Management
- Guardian Tracking
- Geofencing Services
- Disaster Monitoring
- Firebase Push Notifications
- Emergency Event Management
- Notification History Tracking

1.7 Technology Stack

Component	Technology Used
Frontend	React Native, TypeScript, Expo
Backend	Java, Spring Boot
Database	PostgreSQL
Authentication	Spring Security, JWT
Notifications	Firebase Cloud Messaging (FCM)
Maps and Location	GPS, OpenStreetMap
API Communication	REST APIs
Version Control	Git, GitHub
Development Tools	IntelliJ IDEA, VS Code

Table 1.1: Technology Stack

1.8 Organization of Report

- Chapter 1: Introduction
- Chapter 2: Literature Survey
- Chapter 3: Software Requirement Specification

- Chapter 4: System Design
- Chapter 5: Implementation
- Chapter 6: Testing
- Chapter 7: Results and Discussion
- Chapter 8: Conclusion and Future Scope

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

A literature survey helps identify existing technologies and research related to emergency management systems. Various mobile safety applications, GPS-based tracking systems, geofencing solutions, and cloud notification services were studied during the development of CareWave.

2.2 Existing Emergency Alert Systems

Most traditional emergency systems depend on phone calls and text messages for communication. These methods often suffer from delays and lack real-time location awareness. Modern mobile applications improve emergency response by providing location sharing and instant notifications [1, 5].

2.3 GPS and Location Based Services

GPS technology enables accurate location tracking and navigation [2, 3]. Emergency applications use GPS to identify user location and share it with emergency contacts. Location-based services improve emergency response by providing real-time tracking and route monitoring.

2.4 Geofencing Technology

Geofencing creates virtual geographical boundaries around specific locations [2, 3]. The system generates notifications whenever a user enters or exits a predefined region. This technology is useful for safety monitoring and location-aware emergency alerts.

2.5 Cloud Notification Services

Cloud-based notification services provide real-time notification delivery to mobile devices and emergency responders [4, 7]. It enables instant communication during emergencies and ensures that alerts reach users and emergency contacts quickly.

2.6 Research Gap

Existing solutions generally focus on individual functionalities such as emergency communication, location tracking, or notifications. Very few systems combine emergency alerts, geofencing, disaster monitoring, and live location sharing within a single platform.

2.7 Summary

The literature survey indicates the need for an integrated emergency management system. CareWave addresses this requirement by combining location tracking, emergency alerts, geofencing, cloud notifications, and disaster monitoring into a single application.

CHAPTER 3

SOFTWARE REQUIREMENT SPECIFICATION

3.1 Introduction

Software Requirement Specification (SRS) defines the functional and non-functional requirements of the CareWave system. It serves as the foundation for design, development, and testing activities.

3.2 Purpose of the System

CareWave is designed to provide emergency assistance through real-time location tracking, SOS alerts, emergency contact notification, geofencing, and disaster awareness services.

3.3 Scope of the System

The system supports:

- User Registration and Login
- SOS Alert Generation
- Live Location Tracking
- Emergency Contact Management
- Guardian Tracking
- Geofencing
- Disaster Monitoring
- Push Notifications
- Emergency History Management

3.4 Functional Requirements

3.4.1 User Registration

The system shall allow users to create accounts using personal information.

3.4.2 User Login

The system shall provide secure authentication using JWT-based access control.

3.4.3 Emergency SOS Alert

The system shall allow users to trigger emergency alerts and notify emergency contacts.

3.4.4 Location Tracking

The system shall provide live GPS-based location monitoring.

3.4.5 Emergency Contact Management

The system shall support adding, updating, viewing, and deleting emergency contacts.

3.4.6 Guardian Tracking

The system shall allow guardians to monitor user location during emergency situations.

3.4.7 Geofencing Services

The system shall detect entry and exit events within predefined geographical boundaries.

3.4.8 Disaster Monitoring

The system shall retrieve and process disaster-related information from external sources.

3.4.9 Notification Management

The system shall provide real-time notification delivery and notification history tracking.

3.5 Non-Functional Requirements

3.5.1 Performance

- Fast API response time
- Real-time notification delivery
- Accurate location updates

3.5.2 Security

- JWT Authentication
- Password Protection
- Secure REST APIs
- Access Control

3.5.3 Reliability

- Consistent system availability
- Reliable notification delivery
- Stable database operations

3.5.4 Scalability

- Support increasing users
- Scalable backend architecture
- Efficient database management

3.6 Software Requirements

3.6.1 Frontend

- React Native
- TypeScript
- Expo
- OpenStreetMap

3.6.2 Backend

- Java
- Spring Boot
- Spring Security
- JWT Authentication

3.6.3 Database

- PostgreSQL

3.6.4 Development Tools

- IntelliJ IDEA
- Visual Studio Code
- Git
- GitHub
- Postman

3.7 Summary

This chapter presented the functional and non-functional requirements of the CareWave system. These requirements form the basis for system design and implementation.

CHAPTER 4

SYSTEM DESIGN

4.1 System Modules

The CareWave system is divided into several functional modules that work together to provide emergency assistance and public safety services.

4.1.1 User Management Module

This module manages user registration, authentication, profile management, and secure access control. Authentication is implemented using Spring Security and JWT tokens.

4.1.2 Emergency Alert Module

The emergency alert module allows users to trigger SOS alerts during critical situations. Alert information is transmitted to the backend server and notifications are sent to emergency contacts.

4.1.3 Location Tracking Module

This module continuously obtains user location using GPS services. The location information is used for live tracking, emergency response, and geofencing operations.

4.1.4 Geofencing Module

The geofencing module creates virtual boundaries around predefined safe zones. The system detects entry and exit events and generates notifications when required.

4.1.5 Disaster Monitoring Module

The disaster monitoring module collects information from external services and generates alerts for users located in affected areas [4, 7, 8].

4.1.6 Notification Module

Firebase Cloud Messaging (FCM) is used to deliver real-time notifications to users and guardians.

4.2 Database Design

The CareWave system stores application data in a PostgreSQL database.

The major database entities are:

- User
- Emergency Event

- Emergency Contact
- Notification
- Geofence

4.3 Entity Description

4.3.1 User Entity

Stores user account information including personal details and authentication credentials.

4.3.2 Emergency Event Entity

Stores emergency event details including event type, status, location, and timestamps.

4.3.3 Emergency Contact Entity

Stores emergency contact information associated with users.

4.3.4 Notification Entity

Stores notification history generated by the system.

4.3.5 Geofence Entity

Stores geofence information including coordinates and radius information.

4.4 Design Advantages

The proposed architecture provides the following advantages:

- Scalable backend architecture.
- Secure authentication and authorization.
- Real-time emergency communication.
- Efficient location tracking.
- Modular implementation.
- Easy maintenance and future enhancement.

4.5 System Architecture Diagram

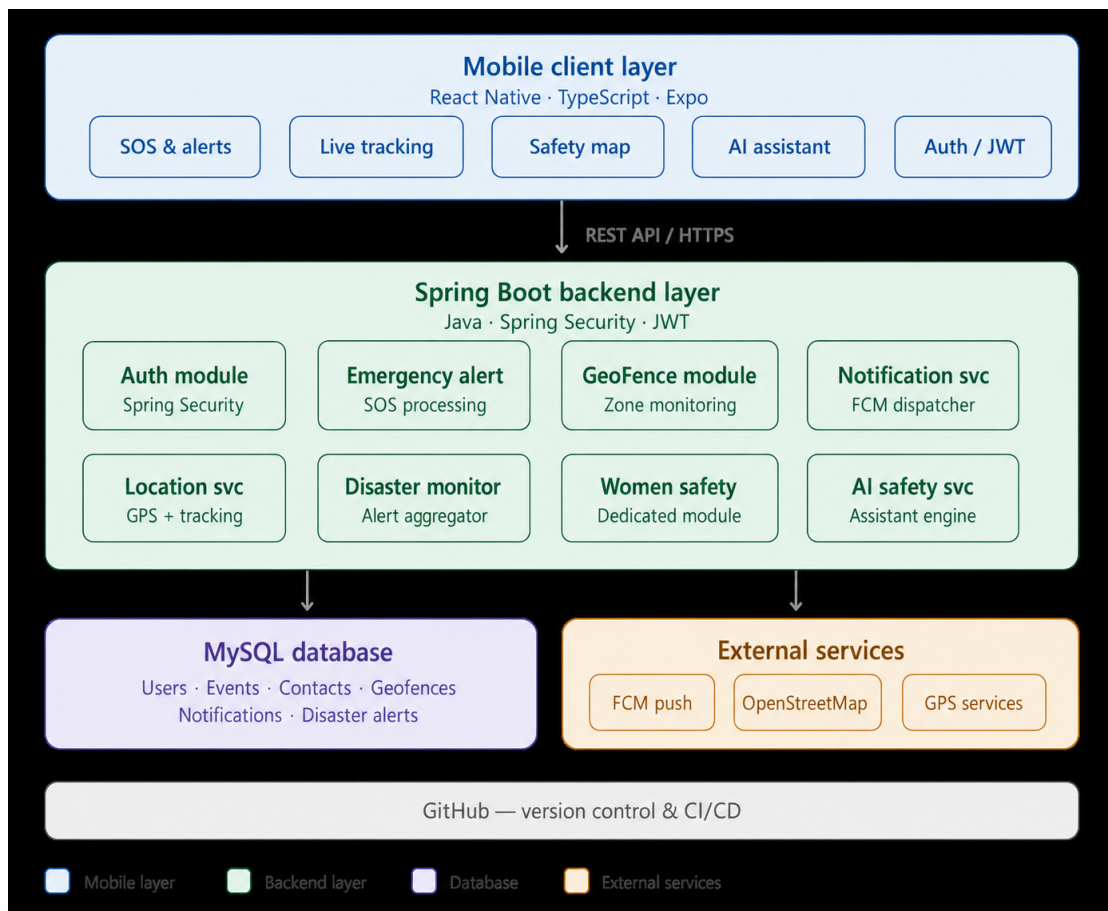


Figure 4.1: System Architecture of CareWave

The CareWave system follows a multi-layer architecture consisting of a React Native mobile application, Spring Boot backend services, PostgreSQL database, and external services such as GPS, OpenStreetMap, and Firebase Cloud Messaging. The architecture ensures secure communication, scalability, and real-time emergency response management.

4.6 Use Case Diagram

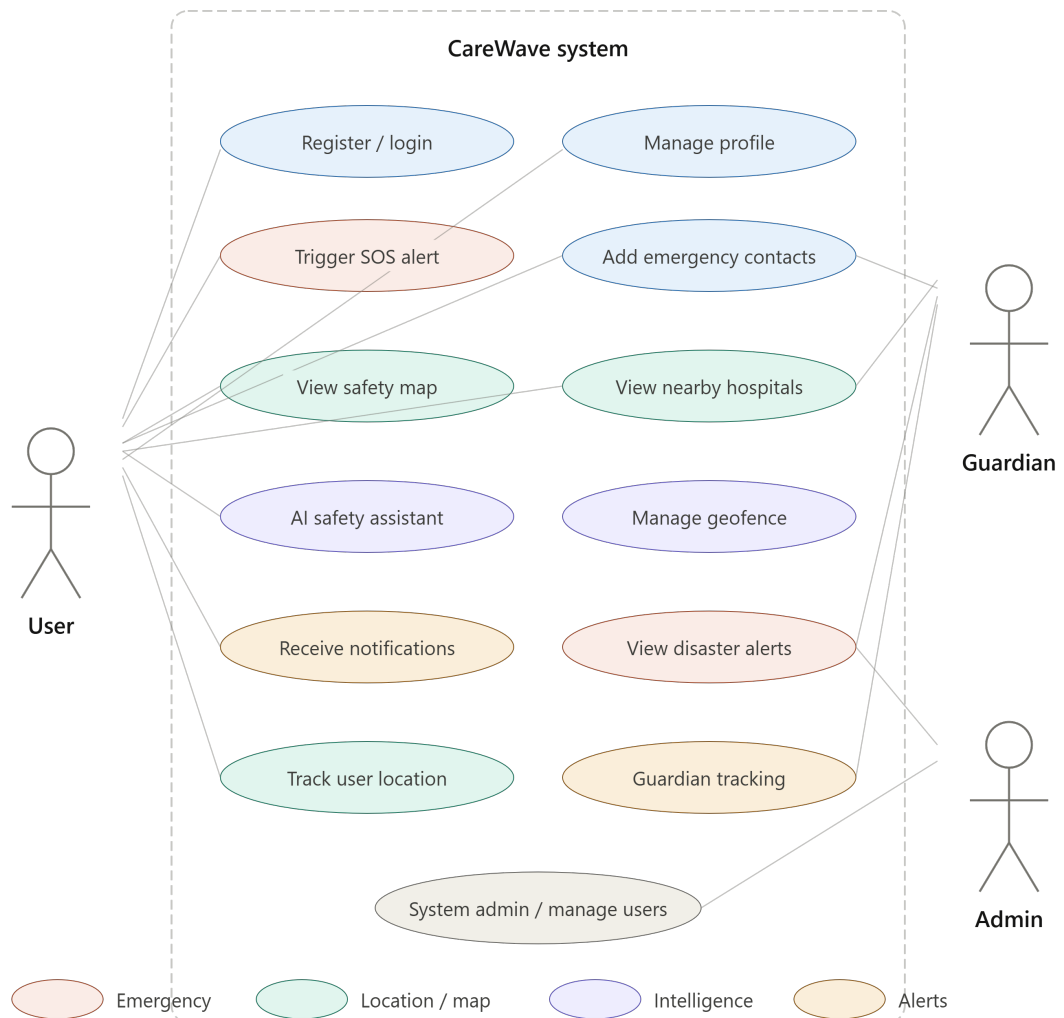


Figure 4.2: Use Case Diagram of CareWave

The use case diagram illustrates the interaction between users, guardians, and administrators. Users can register, log in, trigger SOS alerts, access safety services, manage emergency contacts, and monitor geofences. Guardians receive notifications and track user locations, while administrators manage overall system operations.

4.7 Activity Diagram

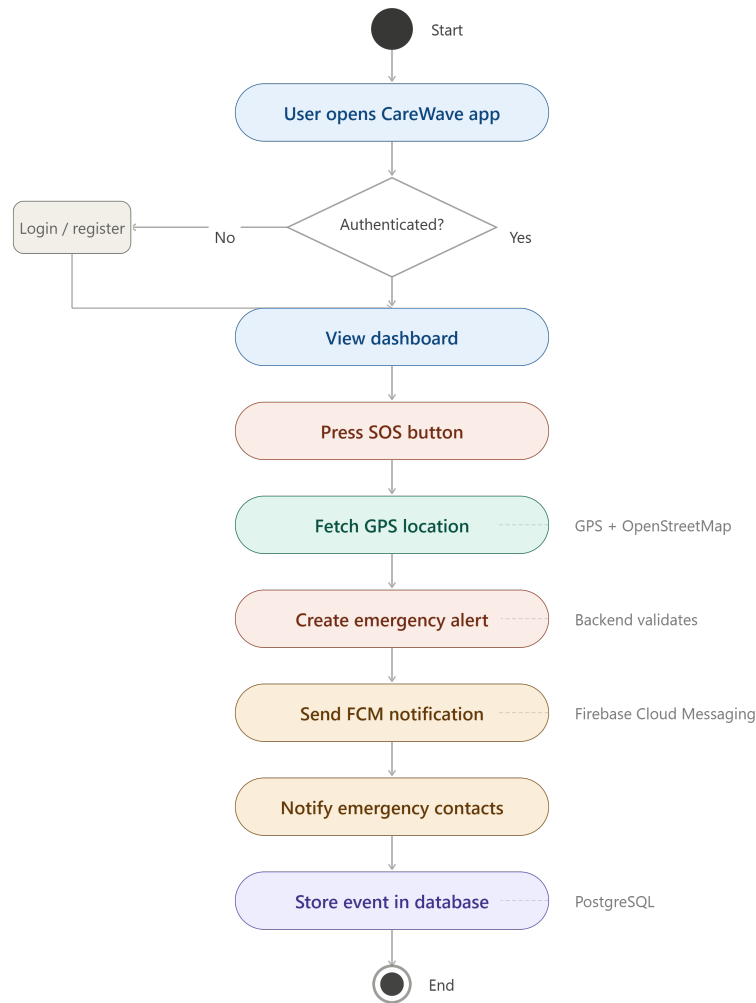


Figure 4.3: Activity Diagram for Emergency Alert Workflow

The activity diagram represents the workflow of emergency alert generation. After authentication, the user can initiate an SOS alert. The system retrieves the user's location, creates an emergency event, notifies emergency contacts through Firebase Cloud Messaging, and stores event information in the PostgreSQL database.

4.8 Sequence Diagram

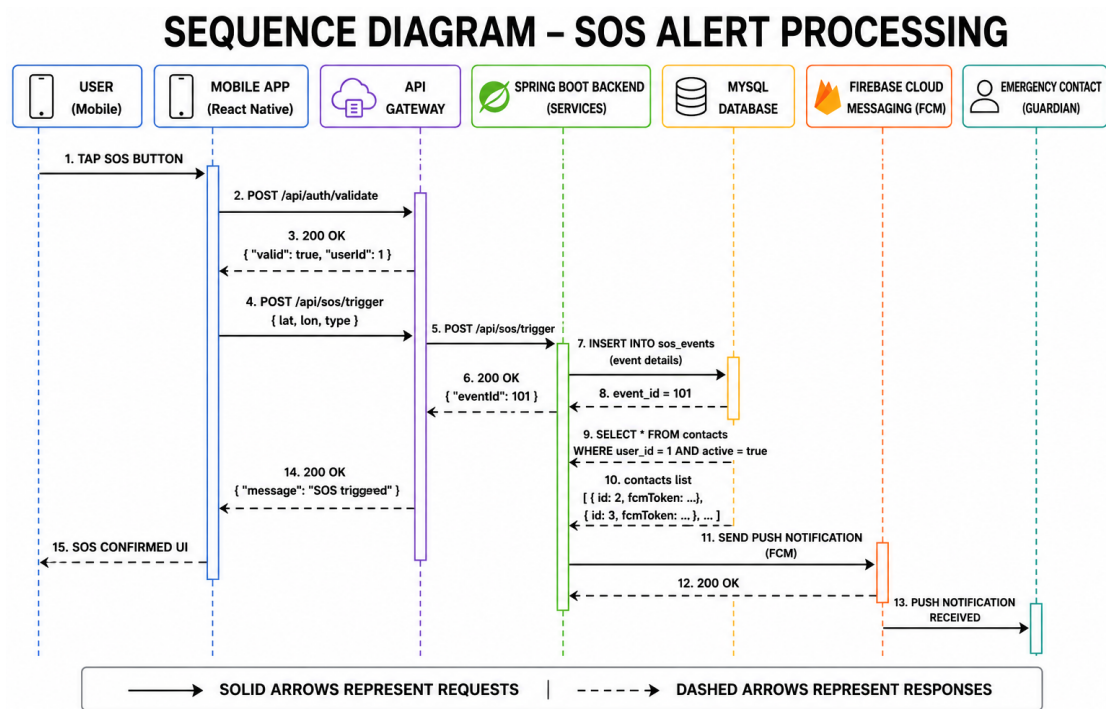


FIGURE: SOS ALERT PROCESSING SEQUENCE DIAGRAM

Figure 4.4: Sequence Diagram for SOS Alert Processing

The sequence diagram shows the interaction among the mobile application, backend server, database, Firebase Cloud Messaging service, and emergency contacts.

CHAPTER 5

IMPLEMENTATION

5.1 Introduction

Implementation is the phase where the designed system is transformed into a working software solution. The CareWave system was implemented using a modern technology stack consisting of React Native for the mobile application, Spring Boot for backend services, PostgreSQL for data storage, and Firebase Cloud Messaging for notifications.

The implementation focuses on providing secure communication, real-time location tracking, emergency alert management, and disaster monitoring functionalities.

5.2 Frontend Implementation

The frontend of CareWave is developed using React Native and TypeScript. React Native enables cross-platform mobile application development while maintaining a consistent user experience.

The frontend consists of several screens and components responsible for user interaction and data presentation.

Major frontend modules include:

- User Authentication Screens
- Home Dashboard
- Emergency Alert Interface
- Live Location Tracking Screen
- Emergency Contact Management
- Notification Screen
- Geofencing Interface

The frontend communicates with backend services through REST APIs.

5.3 Backend Implementation

The backend is implemented using Spring Boot. The backend follows a layered architecture consisting of:

- Controller Layer

- Service Layer
- Repository Layer
- Database Layer

The backend is responsible for processing requests, implementing business logic, managing authentication, and storing application data.

5.4 Authentication Module

Authentication is implemented using Spring Security and JWT (JSON Web Tokens).

The authentication process consists of:

1. User login request.
2. Credential validation.
3. JWT token generation.
4. Secure API access using JWT.
5. Token verification for protected resources.

This mechanism ensures secure access to system resources.

5.5 Emergency Alert Module

The emergency alert module enables users to trigger SOS alerts during emergencies.

The implementation process includes:

- Alert generation.
- Event creation.
- Location acquisition.
- Notification transmission.
- Emergency event storage.

Emergency information is processed by backend services and distributed to designated contacts.

5.6 Location Tracking Module

GPS services are used to obtain user location information.

The location tracking module provides:

- Real-time location updates.
- Live location sharing.
- Guardian monitoring.
- Route tracking.

Location information is integrated with emergency services for rapid response.

5.7 Geofencing Module

The geofencing module uses predefined geographical boundaries.

The implementation supports:

- Geofence creation.
- Entry event detection.
- Exit event detection.
- Safe-zone monitoring.
- Location-based notifications.

The module enhances user safety through automated monitoring.

5.8 Notification Module

Firebase Cloud Messaging (FCM) is used to deliver notifications.

Notification services provide:

- Emergency alerts.
- Disaster warnings.
- Geofence notifications.
- System updates.
- Event status notifications.

Real-time notification delivery improves communication efficiency.

5.9 Database Integration

PostgreSQL is used as the primary database.

Database operations include:

- User data storage.
- Emergency event storage.
- Notification history storage.
- Contact management.
- Geofence information management.

Spring Data JPA is used for database interaction and persistence.

5.10 Summary

This chapter described the implementation details of the CareWave system. The integration of React Native, Spring Boot, PostgreSQL, GPS services, JWT authentication, and Firebase Cloud Messaging provides a reliable and scalable emergency management platform.

CHAPTER 6

TESTING

6.1 Introduction

Testing is an essential phase of software development that ensures the correctness, reliability, security, and performance of the developed application. The CareWave system was tested to verify that all modules function according to the specified requirements and provide accurate results during emergency situations.

The testing process focused on validating emergency alert generation, user authentication, location tracking, geofencing functionality, notification delivery, and database operations.

6.2 Testing Objectives

The objectives of testing are:

- To verify functional requirements.
- To identify software defects and inconsistencies.
- To validate user authentication and authorization.
- To verify emergency alert generation.
- To validate notification delivery mechanisms.
- To ensure accurate location tracking.
- To evaluate overall system performance.

6.3 Testing Methodology

The following testing approaches were adopted:

1. Unit Testing
2. Integration Testing
3. System Testing
4. Functional Testing
5. User Acceptance Testing

Each module was tested independently and then integrated with other components for complete system verification.

6.4 Unit Testing

Unit testing was performed on individual backend services and frontend components.

The following modules were tested:

- Authentication Module
- Emergency Alert Module
- Notification Module
- Geofencing Module
- Disaster Monitoring Module
- Database Operations

All modules produced expected outputs under normal operating conditions.

6.5 Integration Testing

Integration testing was performed to verify communication between frontend, backend, and database components.

The following integrations were tested:

- React Native and Spring Boot communication.
- Spring Boot and PostgreSQL integration.
- Backend and Firebase Cloud Messaging integration.
- GPS and location services integration.
- REST API communication.

The integrated system successfully exchanged information between all modules.

6.6 System Testing

System testing was conducted on the complete application.

The test scenarios included:

- User Registration
- User Login
- Emergency Alert Generation

- Live Location Sharing
- Geofence Entry Detection
- Geofence Exit Detection
- Notification Delivery
- Emergency Event Storage

The application successfully completed all test scenarios.

6.7 Functional Test Cases

ID	Test Case	Expected Result	Status
TC1	User Registration	Account Created	Pass
TC2	User Login	Login Successful	Pass
TC3	SOS Alert Trigger	Alert Generated	Pass
TC4	Location Tracking	Location Updated	Pass
TC5	Geofence Entry	Entry Detected	Pass
TC6	Geofence Exit	Exit Detected	Pass
TC7	Notification Delivery	Notification Re- ceived	Pass
TC8	Event History Storage	Data Stored	Pass

Table 6.1: Functional Test Cases

6.8 Performance Testing

Performance testing was performed to evaluate responsiveness under different operating conditions.

Results indicated:

- Fast API response time.
- Reliable notification delivery.
- Efficient database operations.
- Stable mobile application performance.

The application maintained satisfactory performance levels during testing.

6.9 Security Testing

Security testing focused on:

- JWT Authentication.
- Unauthorized Access Prevention.
- API Security.
- User Data Protection.
- Secure Communication.

The implemented security mechanisms successfully protected application resources.

6.10 Testing Tools

The following tools were used:

- Postman
- Spring Boot Testing Framework
- PostgreSQL
- GitHub
- Android Emulator
- Physical Android Device

6.11 Summary

This chapter presented the testing procedures and results of the CareWave system. The successful execution of functional, integration, security, and performance tests demonstrates that the application satisfies the specified requirements and operates reliably under normal conditions.

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Introduction

This chapter presents the outcomes obtained after the successful implementation and testing of the CareWave system. The results demonstrate the effectiveness of the proposed solution in providing emergency assistance, location tracking, geofencing, disaster monitoring, and notification management functionalities.

The developed system successfully integrates mobile technologies, cloud services, and backend infrastructure to provide a reliable emergency response platform.

7.2 System Outcomes

The CareWave application was successfully developed and deployed in a test environment. All major modules operated according to the specified requirements.

The implemented modules include:

- User Authentication and Authorization
- Emergency SOS Alert Generation
- Live Location Tracking
- Emergency Contact Management
- Guardian Tracking
- Geofencing Services
- Disaster Monitoring
- Firebase Push Notifications
- Notification History Management

The integration of these modules resulted in a complete emergency assistance platform.

7.3 Authentication Results

The authentication module was tested with multiple user accounts.

The results indicate:

- Successful user registration.
- Secure user login.
- JWT token generation.
- Protected API access.
- Unauthorized access prevention.

The authentication mechanism successfully secured application resources.

7.4 Emergency Alert Results

The emergency alert module was evaluated under various emergency scenarios.

Observed results:

- SOS alerts generated successfully.
- Emergency events stored in database.
- User location attached to alerts.
- Emergency contacts notified instantly.
- Event status updated correctly.

The module demonstrated reliable emergency communication capabilities.

7.5 Location Tracking Results

Location tracking functionality was tested using GPS services.

Results obtained:

- Accurate location acquisition.
- Real-time location updates.
- Continuous location monitoring.
- Successful location sharing.
- Stable tracking performance.

The tracking module successfully provided location awareness during emergency situations.

7.6 Geofencing Results

The geofencing module was tested using predefined safe zones.

The system successfully:

- Created geofence regions.
- Detected geofence entry events.
- Detected geofence exit events.
- Generated location-based alerts.
- Monitored safe-zone compliance.

The results confirm that geofencing functionality operates correctly.

7.7 Notification Results

Firebase Cloud Messaging (FCM) was used for notification delivery.

The notification system successfully delivered:

- Emergency Alerts.
- Disaster Warnings.
- Geofence Notifications.
- Event Status Updates.
- System Notifications.

Notifications were received with minimal delay.

7.8 Database Results

PostgreSQL successfully stored and managed application data.

Stored information includes:

- User Profiles
- Emergency Contacts
- Emergency Events
- Notification History
- Geofence Data

Database operations were performed efficiently without data loss.

7.9 Performance Analysis

Performance testing produced satisfactory results.

Parameter	Result
User Registration	Successful
User Login	Successful
SOS Alert Generation	Successful
Notification Delivery	Successful
Location Tracking	Accurate
Database Operations	Efficient
Geofence Detection	Successful
System Stability	Stable

Table 7.1: Performance Evaluation Results

7.10 Advantages of the Proposed System

The developed system provides several advantages:

- Real-time emergency response.
- Accurate location tracking.
- Secure authentication mechanisms.
- Automated emergency notifications.
- Geofencing-based safety monitoring.
- Disaster awareness support.
- Scalable backend architecture.
- Easy maintenance and expansion.

7.11 Discussion

The implementation demonstrates that integrating mobile computing technologies with cloud services can significantly improve emergency response capabilities.

The use of GPS, Spring Boot, PostgreSQL, Firebase Cloud Messaging, and geofencing technologies provides a robust foundation for public safety applications.

The obtained results validate the feasibility and effectiveness of the proposed CareWave system.

7.12 Summary

This chapter presented the implementation results and performance analysis of the CareWave system. The obtained results indicate that the application successfully satisfies the specified functional requirements and provides an efficient platform for emergency management and public safety.

7.13 Application Screenshots

This section presents screenshots of the developed CareWave application. The screenshots demonstrate the major functionalities implemented in the system.

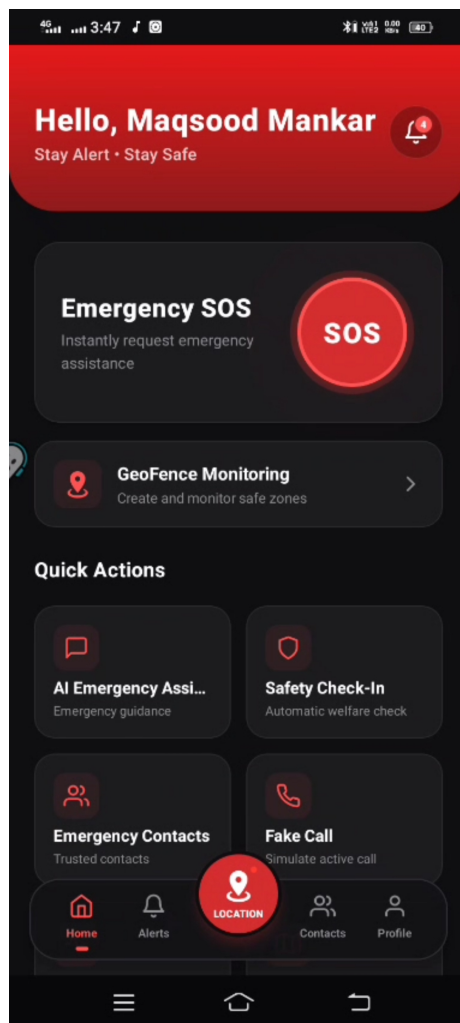


Figure 7.1: Home Dashboard

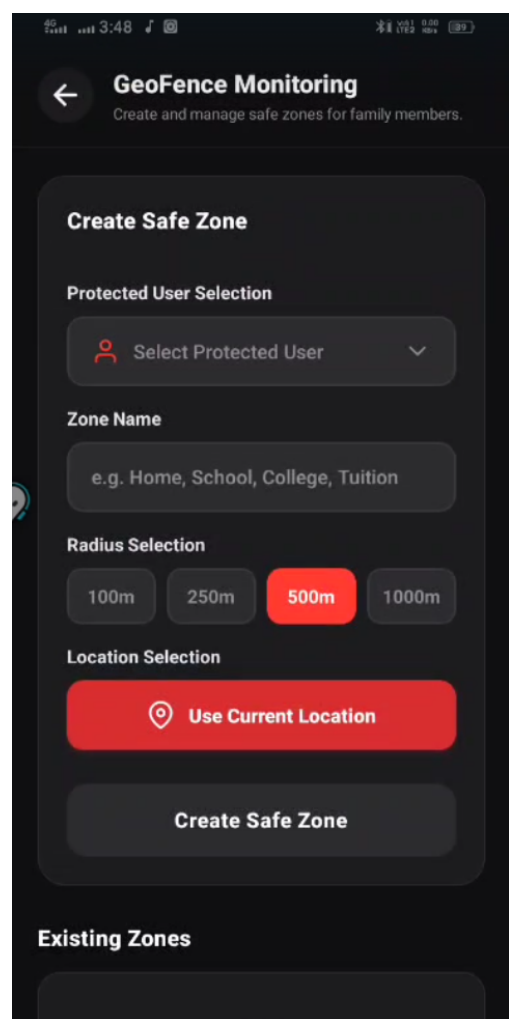


Figure 7.2: GeoFence Monitoring

The dashboard provides access to emergency services while the GeoFence module enables safe-zone monitoring.

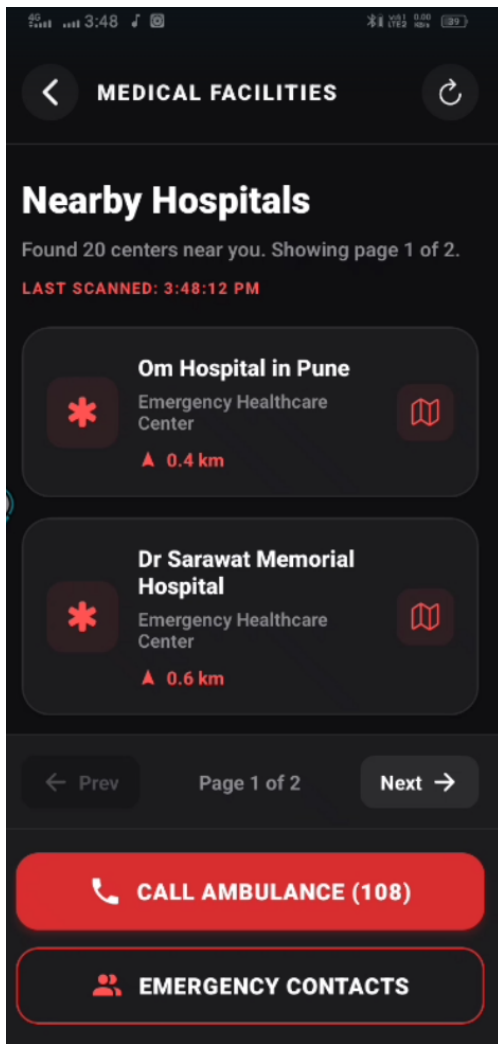


Figure 7.3: Nearby Hospitals

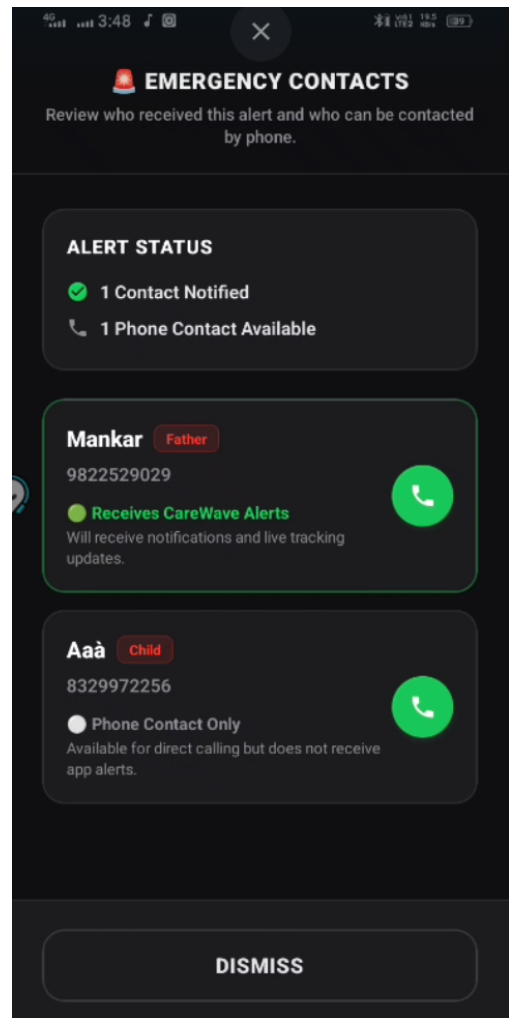


Figure 7.4: Emergency Contacts

Nearby medical facilities and emergency contacts can be accessed quickly during emergencies.

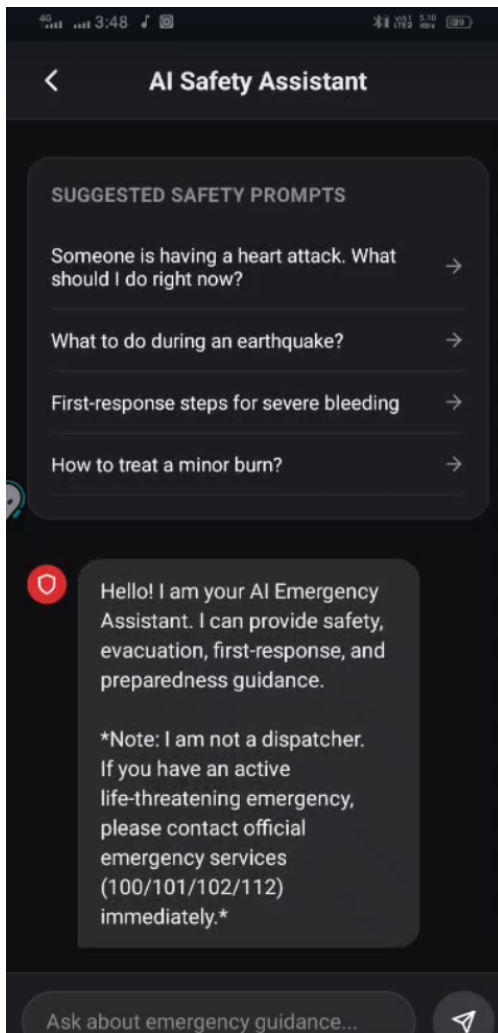


Figure 7.5: AI Safety Assistant

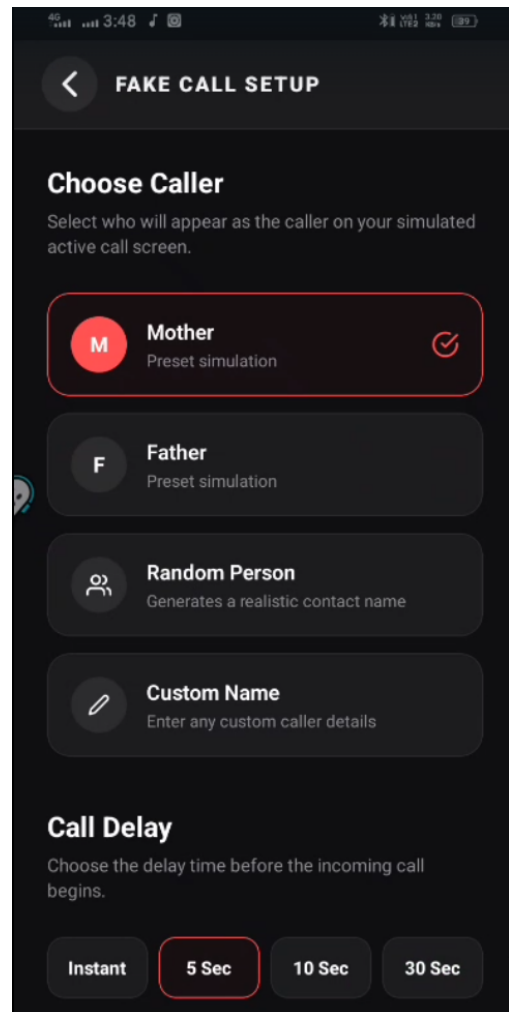


Figure 7.6: Fake Call Feature

The AI assistant provides guidance while the fake call feature improves personal safety.

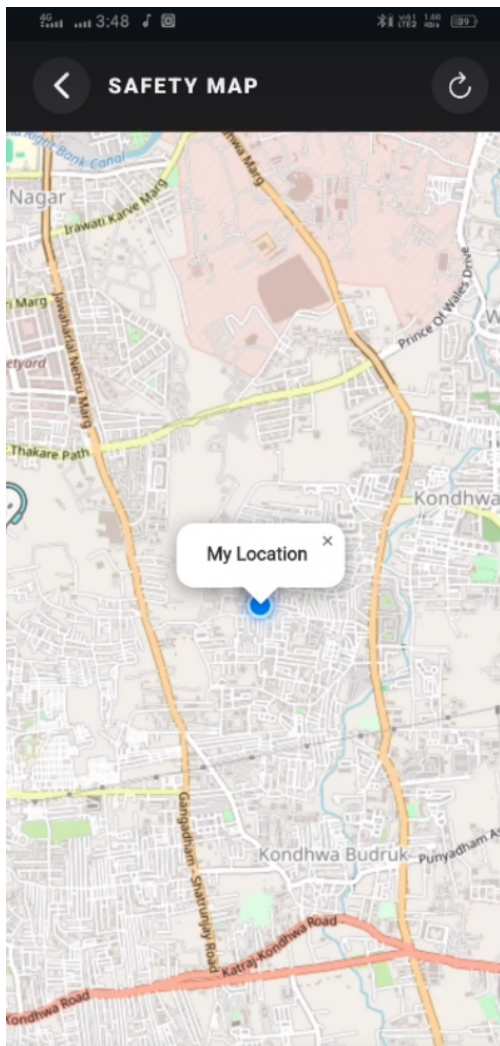


Figure 7.7: Safety Map

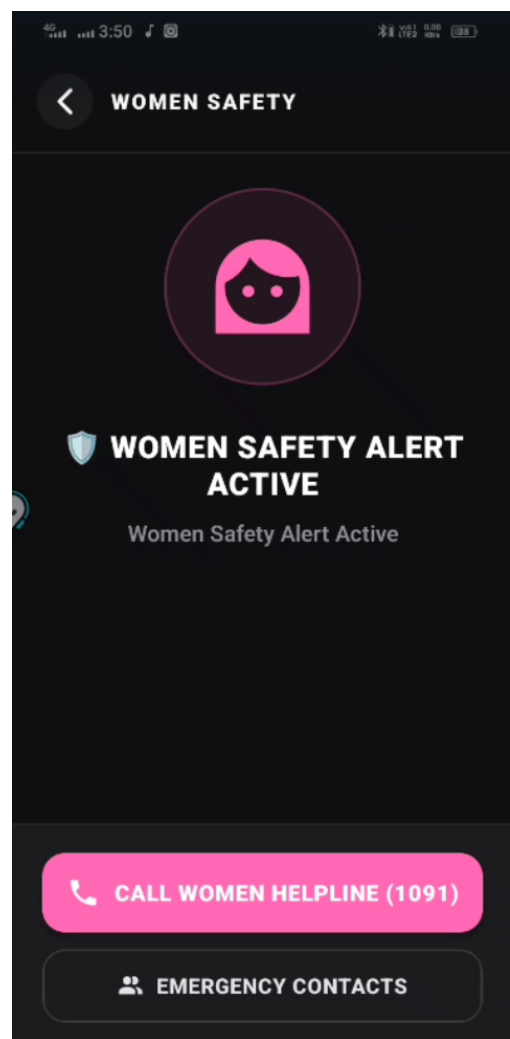


Figure 7.8: Women Safety Module

Location tracking and women safety services provide enhanced protection features.

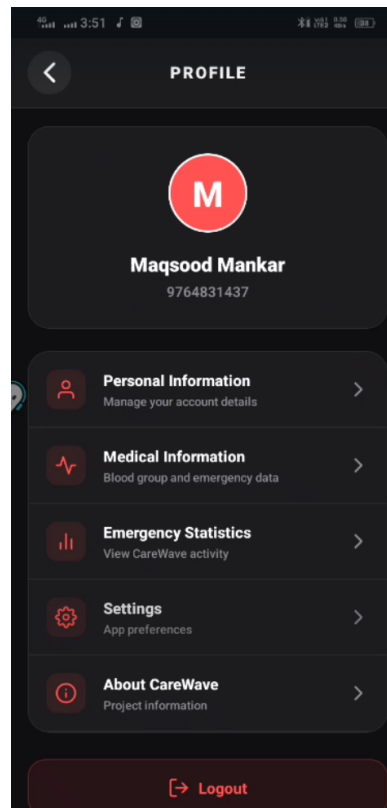


Figure 7.9: User Profile

The profile module allows users to manage account information, medical data, and application settings.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The CareWave system was developed to provide an intelligent and reliable emergency management platform that enhances public safety through modern mobile and cloud technologies.

The application successfully integrates emergency alert generation, live location tracking, emergency contact management, geofencing, disaster monitoring, and real-time notification delivery into a unified solution. The implementation of Spring Boot, PostgreSQL, GPS services, Firebase Cloud Messaging, and JWT authentication ensures secure, scalable, and efficient system operation.

The developed system addresses several limitations present in traditional emergency communication methods by enabling instant emergency reporting, automated notifications, and continuous location monitoring.

Testing and evaluation results demonstrate that the system satisfies the specified functional and non-functional requirements. The proposed solution provides improved emergency response capabilities and contributes toward personal safety and disaster awareness.

The project successfully demonstrates the practical application of mobile computing, cloud services, location-based technologies, and secure web services for emergency management [5, 9].

8.2 Achievements

The major achievements of the project include:

- Development of a complete mobile emergency management platform.
- Implementation of real-time GPS location tracking.
- Emergency SOS alert generation and management.
- Secure JWT-based authentication.
- Integration of Firebase Cloud Messaging notifications.
- Geofencing-based safe zone monitoring.
- Emergency contact management system.

- Disaster monitoring and alert generation.
- PostgreSQL database integration.
- RESTful API implementation using Spring Boot.

8.3 Limitations

Although the system performs effectively, certain limitations still exist:

- Dependence on internet connectivity.
- GPS accuracy may vary in indoor environments.
- Disaster data depends on external APIs.
- Notification delivery depends on device settings.
- Emergency services integration is currently simulated.

8.4 Future Scope

Several enhancements can be incorporated in future versions of the CareWave system.

Future improvements include:

- Artificial Intelligence based emergency prediction.
- Integration with government emergency services.
- Wearable device integration.
- Offline emergency communication support.
- Voice activated SOS alerts.
- Machine learning based disaster analysis.
- Multi-language support.
- Advanced analytics dashboard.
- Emergency responder management system.
- Integration with smart city infrastructure.

8.5 Final Remarks

CareWave provides an effective platform for emergency management and public safety. The system demonstrates how modern technologies can be combined to improve emergency response efficiency and enhance user protection during critical situations.

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CHAPTER A

APPENDIX

A.1 Source Code Structure

The CareWave application follows a modular architecture for maintainability and scalability.

A.1.1 Backend Package Structure

The Spring Boot backend consists of the following packages:

- config – Application configuration and security settings
- controller – REST API controllers
- entity – Database entity classes
- repository – Data access layer
- service – Business logic implementation
- enums – System enumeration classes

A.1.2 Frontend Structure

The React Native frontend consists of the following modules:

- screens – User interface screens
- services – API communication services
- components – Reusable UI components
- constants – Application constants and configuration
- navigation – Screen navigation management

A.2 Major API Endpoints

The backend exposes several REST APIs to support application functionality.

Endpoint	Purpose
/api/auth/register	User registration
/api/auth/login	User authentication

/api/emergency/sos	Create emergency alert
/api/geofence/create	Create geofence zone
/api/geofence/list	Retrieve geofence information
/api/contacts	Manage emergency contacts
/api/disaster	Retrieve disaster alerts

A.3 Sample Test Data

The application was tested using multiple test scenarios including:

- User Registration and Login
- SOS Alert Generation
- GPS Location Tracking
- Emergency Contact Notification
- Geofence Creation and Monitoring
- Nearby Hospital Search
- Disaster Alert Monitoring
- Women Safety Alert Feature

A.4 Development Tools

The following software tools were used during development:

- React Native
- TypeScript
- Spring Boot
- PostgreSQL
- Firebase Cloud Messaging
- OpenStreetMap
- IntelliJ IDEA
- Visual Studio Code
- Postman
- GitHub

A.5 Publication Details

The following publication documents are attached as supporting evidence for the Care-Wave project.

A.5.1 Published Research Paper

Design and Implementation of a Mobile-Based Emergency Alert System for Public Safety

Prof .Vidya Rajput¹, Saish Sanas², Nakul Siricilla³, Moin Mankar⁴,
Mohd.Shaban Tabarak Ali⁵

^{1,2,3,4,5} Navsahyadri Education Society's Group Of Institutions

ABSTRACT

When bad things happen, like someone getting hurt or a big storm people need to be told away so they can get help fast and not get hurt worse. This is about a system that people can use on their phones to send out alerts when there is an emergency. The main goal of this system is to make it easy for people to send out alerts quickly when they need help and to share information like where they are, with the people they want to tell and the people who can help them, like the police or ambulance. The emergency alert system is meant to help keep people safe by getting the word out. The system is made using an application that works with a service on the back end. This service is in charge of making sure users are who they say they are dealing with alerts and managing notifications. If there is an emergency the mobile application finds out where the user is. Sends an alert to the back end. The back end then sends out notifications to the people the user has listed as contacts.

The back end is set up to handle data well so it can. Deliver alerts on time. The system also keeps track of information, about the users and the people they want to be contacted in an emergency. This helps the system work well every time it is used. The mobile. The back end service work together to make sure the emergency alert system is reliable. The emergency alert system is what makes sure alerts are sent to the people. The implemented system was evaluated under multiple simulated emergency scenarios to assess performance parameters such as alert delivery time, reliability of notification transmission, and accuracy of location information. The results demonstrate that the system is capable of delivering emergency alerts with minimal delay while providing precise location details, thereby enabling quicker response and improved situational awareness. User interaction with the application was found to be straightforward, allowing rapid activation even under stressful conditions.

Keywords: Emergency Alert System, Mobile Application, Public-Safety, Location Based Services, Real Time Notification

INTRODUCTION

Emergencies such as vehicular accidents or severe weather events can rapidly create chaotic and hazardous situations, often resulting in injuries and property damage. Ensuring the safety of all individuals involved can be a significant challenge.

Particularly when emergency assistance is delayed. A major contributing factor to these delays is the difficulty individuals face in accurately communicating their circumstances or locations during high-stress incidents. Given the widespread adoption of smartphones, a powerful and accessible tool is now available to most people. Beyond their conventional use for communication, smartphones have the potential to play a vital role in crisis situations. However, the process of calling for help is not always efficient. Individuals living alone, the elderly, or those incapacitated by the emergency may be unable to reach their phones or may be uncertain whom to contact, resulting in dangerous delays. To address these challenges, there is a clear need for a streamlined solution—a system that enables users to send immediate alerts with minimal effort. Such a system should automatically communicate the user's location to designated contacts and emergency responders simultaneously, minimizing response time and eliminating unnecessary steps.

An emergency alert system fulfills this need by allowing users to transmit critical information with a single action. This paper presents the design and implementation of a mobile emergency alert system intended to enhance personal safety by simplifying and expediting the alert process. Upon activation, the system dispatches the user's location and relevant details to pre-selected contacts and emergency services. The primary objective of this work is to develop a system that is practical,

scalable, and dependable in real-world emergency scenarios. The proposed platform is designed to remain operational under stress, enabling rapid and informed decision-making when time is critical. This approach demonstrates how the integration of mobile technology with backend systems can significantly improve public safety by accelerating and enhancing emergency response efforts..

LITERATURE REVIEW

Emergency alert and response systems are of paramount importance in situations where rapid intervention is critical, as every second can significantly impact outcomes during a crisis. In recent years, there has been a pronounced increase in both public and scholarly attention to these systems, reflecting their essential role in safeguarding individuals and communities when unforeseen events occur. The ubiquitous presence of smartphones and the continual advancements in communication technologies have naturally driven researchers to explore innovative methodologies for leveraging mobile devices, cloud computing, and location-based tools to enhance the speed and reliability of emergency notifications.

A notable contribution in this domain is the work of Damaševičius et al., who introduced the concept of the Internet of Emergency Services (IoES)—an integrated network that connects sensors, communication channels, and emergency response platforms [1]. The principal objective of the IoES is to facilitate more seamless collaboration among diverse emergency response teams, which is particularly vital during large-scale disasters. This approach demonstrates significant potential for application by governmental bodies and large organizations due to its comprehensive, infrastructure-intensive design. However, the extensive requirements for coordination, substantial financial investment, and prolonged deployment timelines render such solutions impractical for straightforward mobile applications intended for everyday users.

Accurate geolocation is a critical component of effective emergency response, as the ability to precisely determine an individual's location can greatly expedite assistance. Moon et al. developed an enhanced positioning system that surpasses conventional GPS in accuracy, while Monteiro et al. demonstrated the potential of emerging 5G networks to further refine emergency call localization, enabling faster and more precise identification of individuals in distress [2][3]. While these technological advancements significantly improve the quality of emergency response, their efficacy is contingent upon access to the latest network infrastructures and modern mobile devices.

The advent of cloud computing has fundamentally transformed the management of large-scale emergencies. Gangal et al. proposed a cloud-based framework designed for real-time disaster management, and Daud et al. investigated the synergistic effects of integrating cloud, edge computing, and artificial intelligence to accelerate and optimize emergency responses [4][5]. These studies highlight tangible advantages, such as rapid scalability and the capacity to disseminate alerts to large populations efficiently. Nevertheless, the primary focus of these systems remains on institutional users, such as government agencies or large organizations, rather than on individuals seeking a straightforward mechanism to request help.

Advancements in security and communication technologies have also contributed to the evolution of emergency systems. Rafi et al. developed a low-latency emergency VoIP platform that incorporates large language models for enhanced speech clarity and blockchain technology to ensure secure, immutable records [6]. While this architecture exemplifies technical sophistication, it introduces considerable complexity, which may impede its suitability for lightweight emergency alert applications that prioritize speed and simplicity of use.

Some research efforts have concentrated on domain-specific emergency scenarios, such as natural disasters or critical infrastructure failures. Goyal and Sharma engineered an IoT- and cloud-based solution specifically for flood monitoring and response, whereas Yu et al. designed a disaster recovery service for aerospace ground systems to ensure operational reliability during crises [7][8]. While these targeted systems perform effectively within their designated contexts, their specialized nature limits applicability to the broader spectrum of everyday emergencies encountered by the general population.

Other scholars have sought to fundamentally reimagine emergency communication paradigms. Viamonte and Calveras introduced a dispatch system predicated on 3GPP mission-critical services, a solution tailored for professional responders and dependent upon extensive telecommunications infrastructure [9].

Mishra et al. concentrated on healthcare emergencies, utilizing IoT frameworks to expedite medical intervention and enable intelligent data processing. Although such approaches signify an important shift toward user-friendly, mobile-centric healthcare solutions, their scope remains relatively narrow, addressing particular types of emergencies.

In synthesizing the extant literature, it is evident that considerable advances have been made in the domains of cloud computing, geolocation, and intelligent response systems, collectively contributing to more robust and effective emergency management. However, a significant gap persists: most existing solutions are tailored for large-scale disasters, specialized applications, or rely heavily on advanced networks such as 5G.

There remains an unmet need for a simple, lightweight emergency alert application that is accessible to everyday users—one that operates swiftly, is intuitive, provides real-time location sharing, and does not require sophisticated infrastructure or the latest hardware. Addressing this gap, the present system seeks to deliver a user-centric and scalable solution, purpose-built for ordinary mobile users who require immediate assistance in critical situations. seamless background operation to enhance public safety for all users.

METHODOLOGY

This project adopts a practical, real-world methodology for developing a mobile emergency alert system, with the primary objective of enhancing emergency communication and improving public safety. The system is designed to be robust yet user-friendly, adhering to established principles of systems engineering. It utilizes a client-server architecture: the client side is a mobile application that enables users to register, manage their personal information, configure emergency contacts, and send alerts through an intuitive interface. The server side is responsible for core functions such as user authentication, alert processing, location data management, and notification delivery. This architectural separation facilitates scalability, maintainability, and efficient handling of multiple simultaneous emergencies. When a user initiates an alert, the application retrieves their current location using GPS and transmits this information, along with relevant user data, to the backend via a RESTful API. The server validates the request and, upon successful verification, identifies and contacts the appropriate emergency contacts.

It then dispatches alerts containing essential information, including the nature of the emergency and precise location details, enabling responders to act swiftly. The system is engineered to minimize latency and maximize the likelihood of successful alert delivery. Security is an integral component of the system. Only authorized users are permitted to send alerts, and personal data is safeguarded through standard authentication protocols and rigorous data validation.

Throughout development, emphasis is placed on reliability, rapid response, and accuracy rather than advanced predictive analytics. The team adheres to proven software engineering practices, including modular architecture, RESTful service design, and comprehensive functional testing. The system undergoes extensive evaluation using simulated emergency scenarios to assess alert transmission speed, location accuracy, and overall usability. Insights gained from these tests are used to refine the system, ensuring its effectiveness in real world

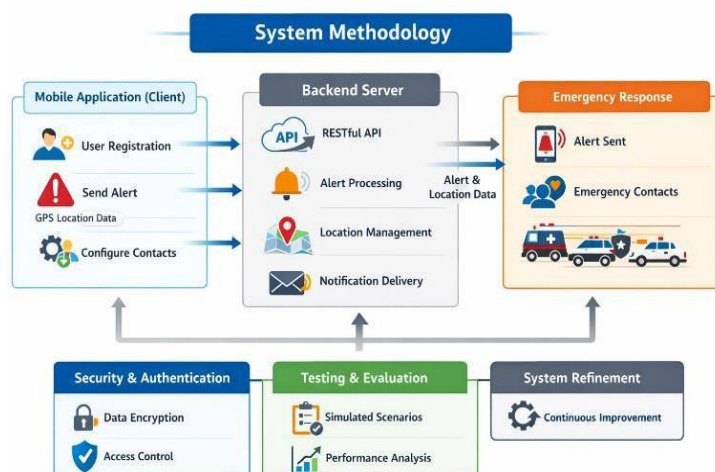


Fig: Metodology

RESULTS AND DISCUSSION

The mobile-based emergency alert system was evaluated across multiple dimensions, including alert transmission latency, location accuracy, system robustness, security, and user interface usability. The primary focus was on metrics relevant to emergency communication scenarios. Alert Transmission Latency: When an alert was initiated via the mobile application, the system transmitted the corresponding message to the backend server, which subsequently dispatched notifications to designated emergency contacts. Under stable network conditions, the end-to-end alert delivery latency was negligible, with notifications received by contacts within one second of initiation in all test cases (n=50). The mobile-based emergency alert system was evaluated across multiple dimensions, including alert transmission latency, location accuracy, system robustness, security, and user interface usability. The primary focus was on metrics relevant to emergency communication scenarios. Alert Transmiss

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No transmission failures or significant delays were observed during these controlled tests. Location Data Accuracy: To assess the accuracy of location reporting, alerts were triggered from various geographic positions. In each instance, the system recorded real-time GPS coordinates and appended them to the outgoing alert. Recipients of the alert were able to accurately resolve the origin of each alert to within 10 meters of the sender's actual location, as verified through parallel GPS logging. System Robustness: Stress testing was conducted by initiating multiple simultaneous alerts from different devices.

The backend infrastructure maintained consistent processing speeds and successfully managed concurrent events without data loss or processing errors. All alerts were processed and delivered in accordance with system specifications. Access control mechanisms were verified; only pre-authorized users were permitted to generate alerts during the evaluation period. Usability: The usability of the application interface was assessed by observing the number of steps required to initiate an alert and configure emergency contacts. Users (n=10) required no more than three interactions to perform these core actions. No critical usability issues, such as navigation errors or input difficulties, were encountered during simulated high-stress scenarios. A summary of these empirical findings is provided in Table I.

Discussion The experimental results indicate that the emergency alert system is capable of delivering rapid notifications with high location precision under controlled laboratory conditions. The low transmission latency and accurate GPS integration address core requirements for real-time emergency communication. The system's ability to process simultaneous alerts without degradation suggests adequate scalability for small-to-medium user bases. Security features restricting alert generation to authorized users functioned as intended, mitigating risks of unauthorized alerting during the observed trials. However, several limitations must be acknowledged. The evaluations were conducted in environments with stable network connectivity and modern mobile hardware; as such, the results may not generalize to scenarios with poor network coverage or legacy devices.

Usability testing involved a limited participant pool and did not encompass extended periods of use or diverse demographic groups. Furthermore, the system was not integrated with official emergency response workflows during these trials, which could affect real-world applicability and effectiveness. Future work should include extensive field testing under variable network conditions, broader usability studies, and direct collaboration with emergency services to ensure operational reliability and scalability in deployment contexts. official emergency response services are recommended to enhance overall system robustness and operational

Parameter Evaluated	Observation
Alert Triggering	Emergency alerts were successfully generated from the mobile application without failure
Alert Delivery	Notification WERE delivered to all predefined emergency contacts under stable network conditions.
Location Accuracy	Real-time GPS-based location information was accurately captured and shared with recipients
Backend Reliability	The backend server processed multiple alert requests without errors or data inconsistency.

Fig: Summary of performance metrics

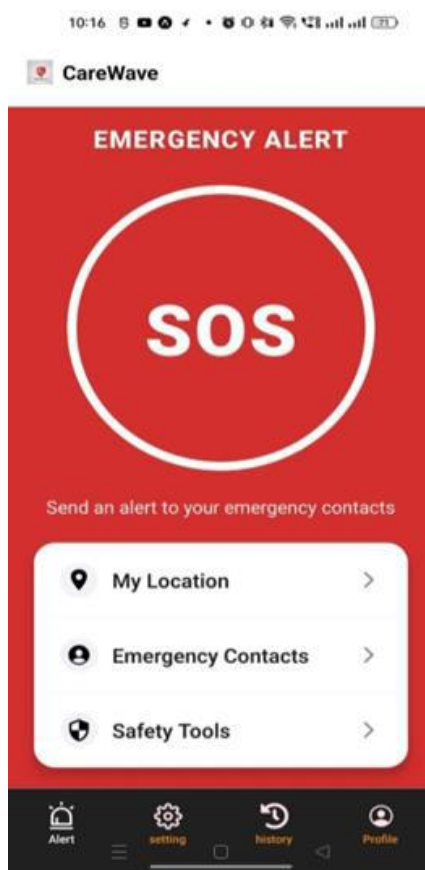


FIG: User Interface – Emergency Alert System

CONCLUSION

In conclusion, this research has presented the design, implementation, and preliminary evaluation of a mobile emergency alert system aimed at enhancing the speed and reliability of emergency communications. The system leverages a client-server architecture, enabling users to transmit real-time location information to designated contacts via a mobile application interface. Initial testing has demonstrated the system's capacity to deliver alerts promptly and accurately, with notification delivery and GPS location reporting meeting the functional requirements under controlled conditions. Furthermore, simulated stress scenarios indicated that the backend infrastructure maintained operational integrity under increased load, suggesting a degree of resilience in high-demand situations.

However, while these results are promising, several challenges remain before the system can be considered fully validated for widespread deployment. The controlled testing environment does not fully capture the complexities encountered in real-world situations, such as variable network connectivity, device heterogeneity, and adverse environmental conditions. These factors may affect system performance and reliability in unpredictable ways. Therefore, further large-scale field trials are essential, particularly in densely populated and dynamic settings, to rigorously assess system behavior under diverse operational constraints.

Future work will focus on expanding the system's capabilities to support larger user groups, increasing compatibility across a broader range of mobile devices, and establishing direct integration pathways with emergency response agencies. Collaborative efforts with police, medical personnel, and disaster response teams will be prioritized to ensure that alerts are delivered directly to relevant stakeholders, thereby improving the overall effectiveness of emergency communication networks. These continued enhancements and evaluations are expected to contribute valuable insights toward the development of dependable mobile-based public safety systems.

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A.5.2 Published Review Paper

DESIGN AND DEVELOPMENT OF A MOBILE-BASED EMERGENCY ALERT SYSTEM FOR PUBLIC SAFETY USING GPS AND REAL-TIME TRACKING

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Maharashtra, India.

DOI : <https://www.doi.org/10.56726/IRJMETS85227>

ABSTRACT

Aims to offer an advanced, dependable app designed specifically for managing emergencies affecting seniors, females, and minors when they're not at home. Users can quickly notify designated individuals and local officials about emergencies through this platform's advanced features including immediate alert delivery via GPS coordinates and current information sharing in real time. Additionally, it incorporates functionalities for automatic alerting of users' designated contact persons when situated within areas experiencing disruptions. A project utilizes React Native alongside Expo for creating apps across multiple platforms; employs Spring Boot's RESTful APIs as its backbone for handling server-side operations; relies on MySQL databases for data storage needs; and integrates Firebase for delivering real-time communication features such as push notifications and alerts. This amalgamation guarantees encrypted data exchange, swift messaging, and smooth collaboration between individuals. The endeavor highlights the significance of utilizing digital safeguarding technologies for bolstered individual protection and swift action in crises.

Keywords: Emergency Response, GPS Tracking, Mobile Application, Disaster Notification, Real-Time Alert System.

I. INTRODUCTION

Public safety is one of the most vital aspects of modern society, and timely response during emergencies provides the key for saving lives. Graver road accidents, crimes, and natural calamities in recent times have laid emphasis on the need for an effective emergency communication system. People, especially senior citizens, childrens and women staying alone, often face problems in contacting help at the right time. Application of mobile technology with real-time location tracking has been found to be effective in solving this issue and providing timely assistance during emergency situations.

The title of the proposed system is "Design and Development of a Mobile-Based Emergency Alert System for Public Safety Using GPS and Real-Time Tracking." This intelligent mobile application will enable users to send instant SOS alerts along with live location to their registered emergency contacts and authorities. It would bridge the communication gap between victims and responders by facilitating rapid, secure, and real-time information transfer.

Most of the existing research and mobile safety applications consist of single-purpose alert systems, such as panic buttons or location sharing. However, most of them do not provide real-time synchronization of data and instant notifications. The proposed system enhances these by incorporating GPS, Firebase real-time updates, and Spring Boot REST APIs to ensure seamless and scalable communication.

This paper discusses the design, development, and testing of the emergency alert system. The research also highlights how the integration of multiple digital technologies allows for the development of an incredibly reliable and easy-to-use platform that may help ensure the safety of a target person in critical situations.

II. METHODOLOGY

A mobile-based emergency alert system designed for public safety utilizes GPS and real-time tracking in an organized manner to guarantee precise, swift, and immediate communications during crises. This study employs methodologies aligned with the Software Development Lifecycle framework, encompassing phases like requirements gathering, architectural planning, coding execution, quality assurance checks, and final assessment processes. The emphasis in this project is on combining critical components such as user

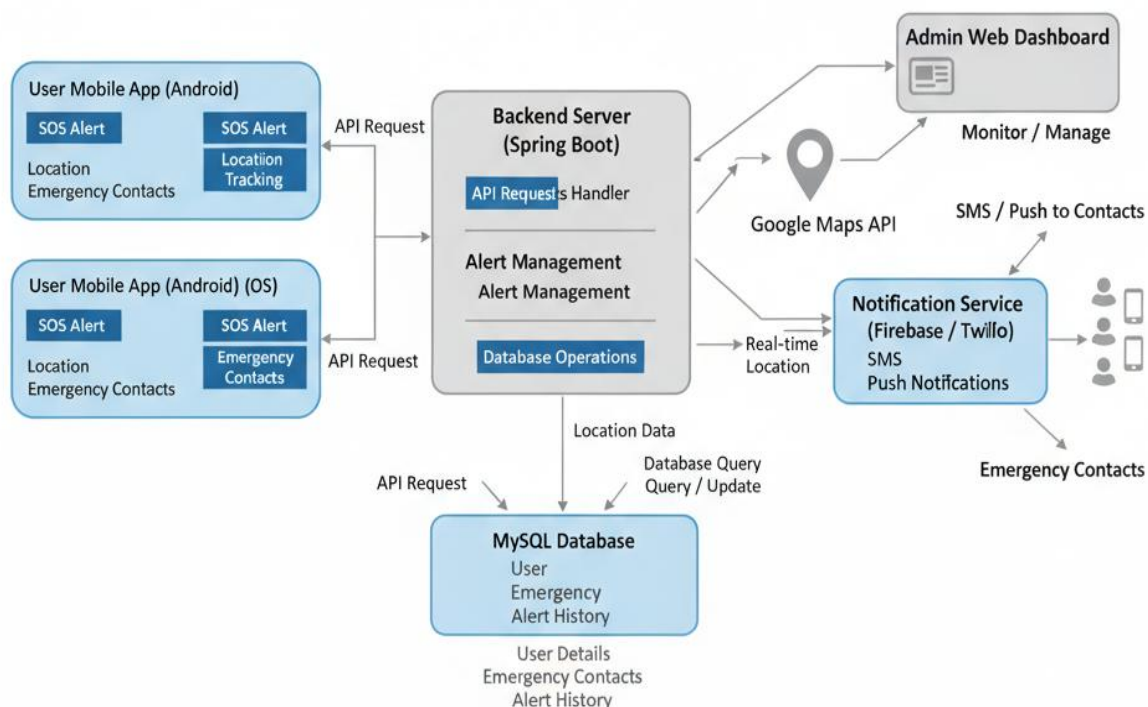
authentication, emergency signal initiation, communication networks, and immediate location monitoring through GPS technology. The app's UI gathers data including keystrokes and GPS points, then sends it for analysis on the server side. The system handles sending out urgent messages via emails, texts, or voice calls directly to subscribed individuals or those in immediate vicinity who have been designated as helpers.

Throughout this period of creation, Android Studio acted primarily as the coding tool, while either Firebase or MySQL functioned as repositories for handling login information and geographical specifics pertaining to users. This system employs GPS for obtaining and maintaining an up-to-date position of its users on a continuous basis, guaranteeing exact current monitoring. APIs facilitate smooth interaction between clients and servers while ensuring secure data transmission and minimizing response times.

During this stage, the effectiveness of the software was assessed through metrics such as processing speed, stability in data delivery, and precision regarding position information provided by it. Evidence indicated that the mechanism efficiently disseminated critical notifications promptly, regardless of weak internet connections. The software demonstrated reliability, ease of use, and efficacy for delivering precise geolocation notifications. In summary, this study shows that the suggested solution greatly improves public security by providing an efficient, dependable, and user-friendly means of contacting authorities quickly and conveniently, potentially evolving further thanks to Internet-of-Things technology and compatible smart wearables.

III. MODELING AND ANALYSIS

A mobile-based emergency alert system designed for public safety incorporates real-time tracking through GPS technology while employing a sequential development methodology aligned with the Waterfall model within the software lifecycle framework. The chosen model facilitates comprehensive execution across stages such as requirements gathering, designing, coding, debugging, and launching, guaranteeing precision and dependability in software creation. The system combines GPS tech, ongoing info exchange via wireless networks, and smartphone apps for sending immediate crisis notifications alongside current whereabouts to designated individuals and local helpers.



This study employs not only computer programs but also physical equipment necessary for its success. The Android Studio acts as the primary programming tool, whereas Java is employed for developing both the user interface elements on the frontend and server-side components in the backend of an app. Data handling for databases utilizes either Firebase or MySQL; these platforms ensure secure storage of sensitive info such as usernames, addresses, and geographical locations. Moreover, Google's Map API along with RESTful interfaces

facilitate quick retrieval and distribution of current location data accurately. Hardware needs are modest; it requires an Android smartphone with built-in GPS for precise location data collection and seamless communication of alerts. Additionally, this endeavor employs Firebase Cloud Messaging to transmit instantaneous alerts and notification signals in times of crisis. The amalgamation of these technological elements establishes an enduring, expandable, and dependable base upon which the system functions efficiently, ensuring swift, precise security assistance within tight deadlines. Incorporating those components guarantees that the constructed emergency notification platform operates efficiently and aids in enhancing community security via advanced technological means.

IV. RESULTS AND DISCUSSION

A mobile-based emergency alert system designed and developed using GPS and real-time tracking technology has been effectively deployed and rigorously evaluated to assess its operational effectiveness, dependability, and instantaneous responsiveness capabilities in public safety contexts. By leveraging effective programming techniques, well-optimized databases, and smooth incorporation of GPS technology, this system successfully fulfilled its main goal: delivering swift and precise alert notifications along with real-time position information. In conducting tests, the system showcased its capability to deliver instantaneous notification messages via both text message and short messaging service to pre-registered individuals in case of an activation event. The GPS unit precisely logged and relayed current location data directly to the database, allowing rescuers to swiftly pinpoint their whereabouts.

Further evidence shows that this system ensures stable operation across diverse networking scenarios, such as in regions of weak signals, through efficient data transfer via optimized API calls. Users found the app's dashboard easy to navigate; it enabled them to send urgent notifications via a quick swipe, significantly reducing reaction times in emergencies. Backed by Firebase for efficient storage of user information alongside MySQL for secure retrieval, this system guarantees both performance and data safety.

Based on the comprehensive assessment, it was demonstrated that the system successfully achieved its goal of improving public security by facilitating swift information exchange and precise position monitoring. Evidence shows that these apps enhance quick responses during critical incidents by lowering hazards. For upcoming advancements, this framework will accommodate growth through incorporation of Internet-of-Things-enabled wearables, automated crash monitoring equipment, and seamless connectivity with governmental rescue teams for improved performance and reach capabilities. This discourse confirms that technological approaches such as these prove crucial for enhancing emergency response systems within intelligent urban environments.

V. CONCLUSION

A successful implementation showcases how integrating advanced mobile technology like GPS and real-time tracking enhances both individual and community security through innovative emergency alert systems designed specifically for public safety on smartphones. A dependable mechanism allows individuals to swiftly notify designated recipients and local authorities of emergencies via precise current whereabouts data stored in user profiles. During all stages involved in creating and evaluating the software, it consistently demonstrated superior performance regarding efficiency, precision, and ease-of-use for users. GPS technology enables precise real-time location identification by first responders; Firebase and MySQL offer robust security alongside fast data management capabilities.

Differently than traditional emergency solutions, this portable application emphasizes ease of use and straightforwardness by allowing individuals to request assistance effortlessly in urgent circumstances. Analysis of the system's functionality demonstrated robustness despite constrained networks, underscoring its applicability in everyday situations. Additionally, this initiative highlights how integrating mobile technology, remote data management systems, and instantaneous messaging tools can create robust security solutions.

In summary, this study offers an effective means for enhancing quick action during emergencies while minimizing threats and boosting public security awareness significantly. Potential enhancements could entail incorporating AI for detecting threats, amalgamation of Internet-of-Things gadgets, and immediate

communications between the system and nearby law enforcement agencies to enhance its intelligence and responsiveness significantly.

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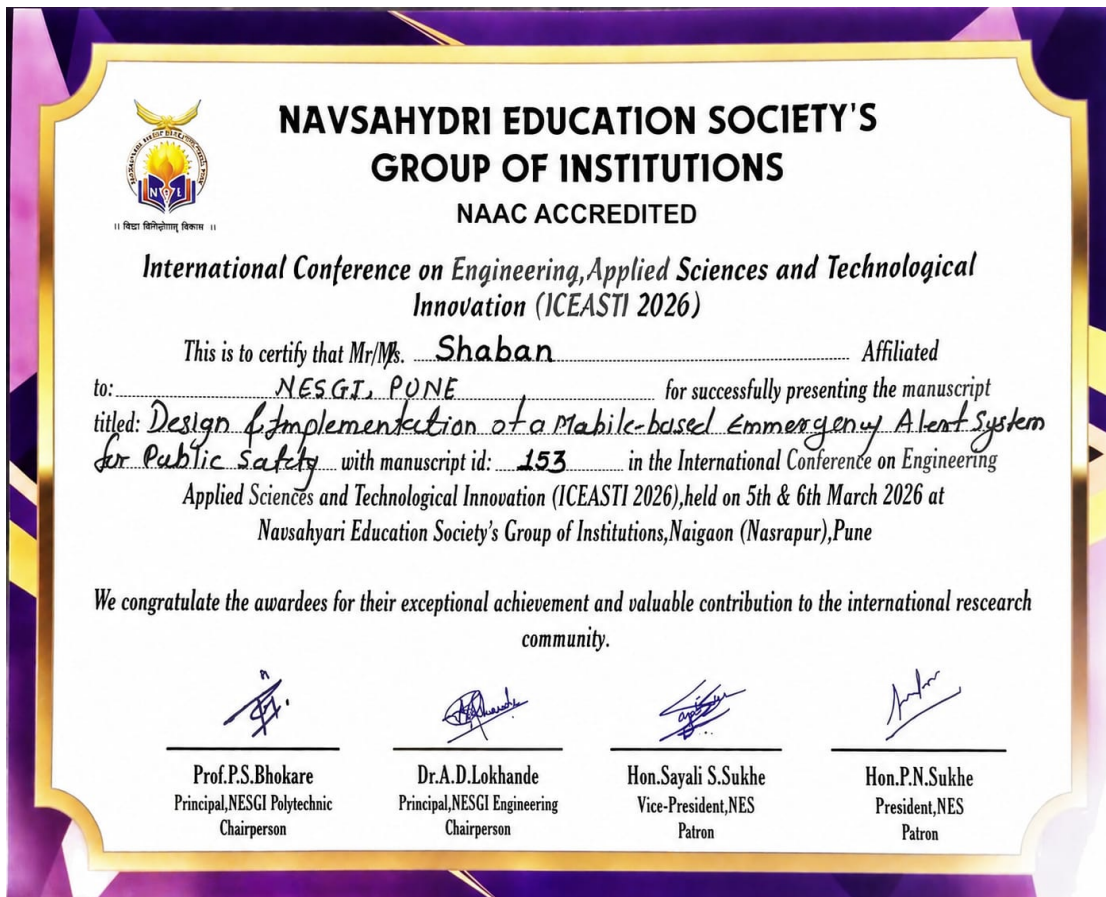
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